



Hack Away Hunger 2026

dsmHack x Corteva

Submission

Count Me In

Volunteer hours, verified — SNAP kept

80 h

Monthly rule

8

Languages

1 sec

To verify

\$303k

Aid preserved / yr

Live app	https://count-me-in-iowa.netlify.app
Live demo stage	https://count-me-in-iowa.netlify.app/stage
Verify an attestation	https://count-me-in-iowa.netlify.app/verify (no key needed)
Source code (MIT)	https://github.com/chessoren/hackawayhunger
Team	Oren Vidal
Prizes	Grand Prize · Community Impact Award · People's Choice Award

We don't create food. We stop it from disappearing.

Submission at a glance

Volunteering → verified hours → SNAP kept. An SMS & voice assistant in 8 languages helps Iowans meet the new 80-hour rule, and gives pantries the regular volunteers they need.

The problem

Since the July 2025 federal law (P.L. 119-21), adults 18–64 without a child under 14 must prove 80 hours a month of work, training or volunteering to keep SNAP; exemptions for veterans, people experiencing homelessness and former foster youth were removed. Iowa SNAP recipients fell 9% in a year (271,880 → 247,907) while 1 in 8 Iowans face food insecurity. The law already counts volunteering hour for hour — **what's missing is the proof.**

What we built

Person

SMS / voice / web assistant in 8 languages: letter photo → 3-sentence summary + deadline; screening decided by a cited, deterministic rules engine; 80-hour gauge; 3 accessible missions; QR or 4-digit check-in; consent-gated sending; DELETE by SMS.

Host site

10-minute onboarding, QR poster, first name only, one-tap validation signed with Ed25519, grant-ready report, audit log.

State

Monthly PDF attestation with QR; verified in one second; any modification breaks the signature.

Community

Anonymized dashboard ($k \geq 10$), map of opportunity deserts for 99 counties, editable impact model ($\approx$ \$303,000/year for 500 people).

How to test in 2 minutes

1. Open **count-me-in-iowa.netlify.app/stage** and paste a Claude or OpenAI key — it stays in your browser.
2. Text **Hi** (or Hola, Jambo), then tap **Send the sample State letter**.
3. Answer in your own words; pick mission 1, 2 or 3.
4. Demo clock: **Shift starts** → **Scan the QR at the door** → **Shift ends** → scan again → **Confirm**.
5. Text **ATTESTATION**, open it, then visit **/dashboard**.

API keys — temporary.

For the submission, the app runs on each user's own AI key; no key is shipped in the code. **If Count Me In wins, the prize funds the AI, SMS and voice keys so the service runs for everyone.** The demo video was recorded with pre-written AI responses; the live app uses the real APIs. Sites, people and the sample letter are illustrative.



Count Me In

Volunteer hours, verified — SNAP kept

• Live in Iowa

80-hour rule

8 languages

80 h

🕒 Monthly rule

1 sec

🛡️ To verify

3

👥 Winners per hour



Count Me In

(515) 800-8080 · English



Hi

Hi! I'm Count Me In. I help you keep your food assistance (SNAP) under the new 80-hour rule — for free, by text or by phone call.

Private by design: I only keep your first name, phone, language and hours. Nothing goes to the State without your YES. Reply DELETE anytime to erase everything, or HELP to talk to a person. I speak: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

What's your first name?



Text message



Send the sample State letter



STATE DEPARTMENT OF HUMAN SERVICES
Food Assistance Program - Field Office 77
P.O. Box 0000, Des Moines, IA 50300

SAMPLE
NOTICE

Date of notice: **October 1, 2026**
Case number: XX-XXXXXX

LINDA M.
1400 E Walnut St Apt 2
Des Moines, IA 50316

NOTICE: ABLE-BODIED ADULT WORK REQUIREMENT

Federal law (Public Law 119-21) changed the rules for Food Assistance (SNAP). Our records show you are between 18 and 64 years old and you do not have a child under 14 in your food assistance household.

Beginning with the benefit month of **November 2026**, you must work, volunteer, or take part in a work or training program for at least **80 hours each month**, unless you qualify for an exemption.

WHAT YOU MUST DO: Within **21 days of the date of this notice**, report your hours or ask for an exemption. Return the enclosed form 470-XXXX with proof of hours worked or volunteered.

If you do not meet the requirement, you can receive food assistance for only **3 months in a 36-month period**.

You may be exempt if you are medically certified as physically or mentally unfit for work, pregnant, or caring for an incapacitated person. Volunteer work with a public or nonprofit organization counts toward the 80 hours when it is verified.

Questions? Call the Contact Center at **1-800-000-0000**, Monday to Friday, 8:00 a.m. to 5:00 p.m. You have the right to appeal this decision.

SAMPLE DOCUMENT FOR DEMONSTRATION — Count Me In, Hack Away Hunger 2026. Not an official notice.

Meet Linda

Linda, 58

Des Moines • cleans houses 10 h a week

• Gets SNAP

No car

80 h

📅 Now required

21 days

🕒 To respond

A two-page letter in administrative English. Without help, by spring she is in the line at a pantry that already breaks records.

What changed in 2025

A federal rule just reached hundreds of thousands of lowans



18 → 64

The 80-hour rule now covers adults up to 64



Child under 14

Parents of teenagers are no longer exempt



Exemptions removed

Veterans, homelessness, former foster youth



3 months in 36

Without 80 h or an exemption, SNAP stops

-9%

Iowa SNAP recipients in one year
271,880 → 247,907

1 in 8

Iowans food insecure
73M-meal gap

The real problem

Not willingness. Proof.

01

They don't know if they're covered

or already exempt

02

They can't read the letter

long, English, tight deadline

03

They can't reach hours

no car, irregular shifts

04

They can't prove what they do

volunteering leaves no trace

05

They miss the deadline

one missed report cuts the aid



The other half

Pantries chronically lack **regular weekday volunteers** — and track attendance on paper they can't certify.

SNAP provides ~9 meals for every 1 from food banks. **Every person who loses SNAP lands in the pantry line.**

• The insight • 7 CFR 273.24

The law already counts volunteering. We built the missing proof.



The person

Keeps SNAP — no app, no password

Keeps food aid



The pantry

Gains regular volunteers

Gains volunteers



The State

Gets clean, signed proof

Clean proof

One hour of volunteering serves three people at once.



Count Me In

(515) 800-8080 · English



📄 SNAP work requirement notice —
State Department of Human Services
(sample)

- Starting in November, you must do 80 hours a month of work, volunteering or training to keep your food assistance.
- If you do not, your food assistance can stop after 3 months.
- Volunteering at a nonprofit counts, if your hours are verified.

🕒 **Deadline: Thu, Oct 22**

✅ Report your hours or ask for an exemption by October 22, with the form and proof of your hours.

📞 1-800-000-0000

I set a reminder 3 days before the deadline.

Let's check together if this rule applies to you. What's your first name?



Text message



Send the sample State letter



01

The letter, decoded

Snap a photo. Get 3 sentences.

Text "Hi" to a number — SMS, voice or web. Photograph the State letter, even crooked. Back, in her language: **what it says, the deadline, the one thing to do**, and a reminder 3 days before.

3

📄 Sentences

8

🌐 Languages

0

📱 Apps

02

Does this apply to me?

6 questions. A cited rule.

She answers in her own words — *"My knees hurt a bit but I can still work."* The AI turns words into facts; a **deterministic rules engine decides** and cites the regulation. Unsure? A human navigator.

1

Question at a time

0

AI decisions

1 file

Per state



Count Me In

(515) 800-8080 · English



About how many hours a week do you work for pay? (Send 0 if none.)

I clean houses, about 10 hours a week

The 80-hour rule applies to you. Don't worry: work, training and volunteering all count, in any mix. I will help you get there.

Rule cited

7 CFR 273.24 — Time limit for able-bodied adults (work = paid, in-kind, unpaid or volunteer work)

Rules file IA 2026.10-hackathon

Your paid work gives about 43 hours a month. Do you have training or English class hours this month? Send the number (0 if none).

0



Text message



Send the sample State letter





Count Me In

(515) 800-8080 · English



example: "weekday mornings" or
"Tuesdays after 2pm".

I am free anytime

I clean houses Monday and Wednesday
mornings

Here are missions that fit you:

1 Food distribution — Grace Church
Community Grocery

Tue, Oct 13, 09:00–13:00 (4 h) · 12 min
walk

2 Food distribution — Eastside
Neighborhood Pantry

Sat, Oct 10, 08:00–12:00 (4 h) · 19 min
walk

3 Welcome desk — Downtown
Community Kitchen

Sun, Oct 11, 12:00–15:00 (3 h) · Route 4
— E 14th St, about 33 min · seated work
Reply 1, 2 or 3.

1

2

3



Text message



Send the sample State letter



03

Three missions, never more

Ranked by real access.

A 12-minute walk or a DART bus line, around her cleaning schedule, optimized to reach **80 hours before the deadline**. She replies "1". At the door: scan the QR, or text a 4-digit code from any basic phone.

3

✓ Choices max

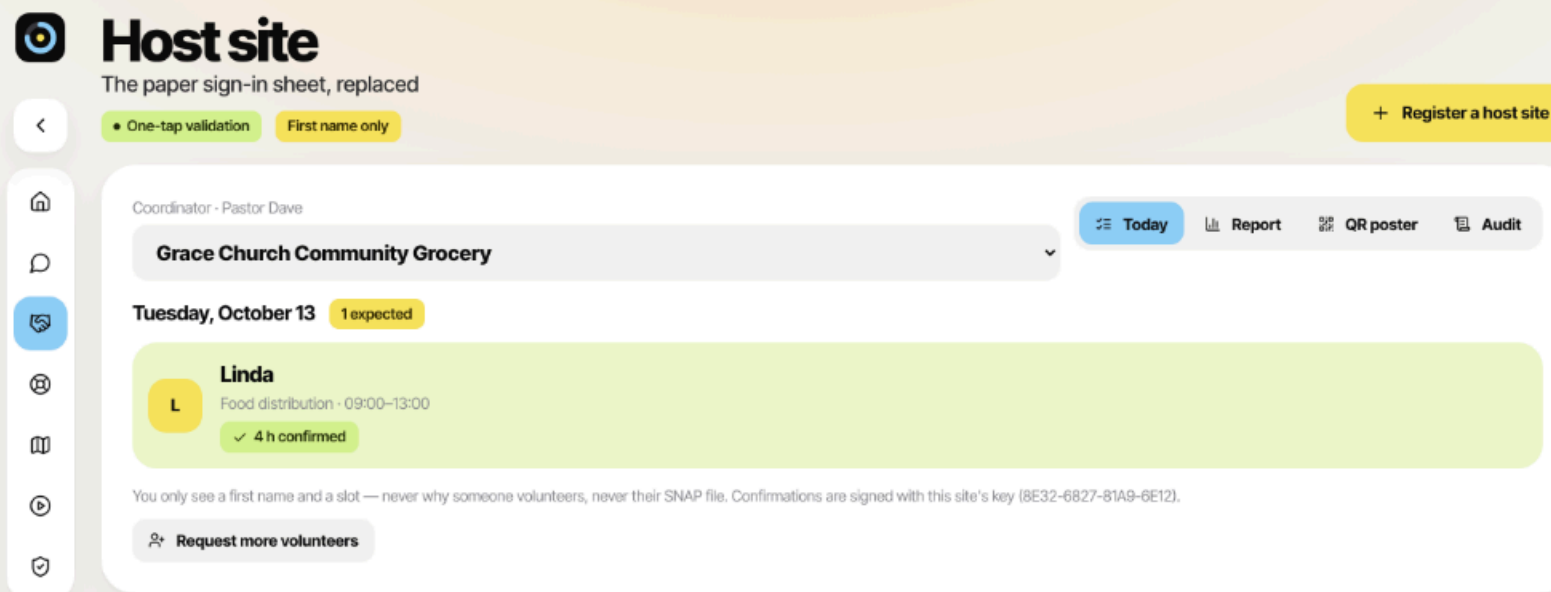
12 min

🚶 Walk

4 digits

☎ Basic phones

The paper sign-in sheet, replaced



10 min
to become a host site



One tap
confirmation, Ed25519 signed



First name
never the SNAP file



Grant-ready
monthly report & CSV

Verified in one second. Forged in none.



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Authentic

Not modified since it was signed by Count Me In.

Linda 2026-10 47 / 80 h

Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	4 h	Pastor Dave key B:312-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 4 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Download the PDF

Verify another document

• **Authentic** signed by the platform key



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Not valid

Signature does not match the document. It was modified after signing, or not signed by Count Me In.

Linda 2026-10 51 / 80 h

Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	8 h	Pastor Dave key B:312-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 8 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Verify another document

The attestation is signed with the Count Me In platform E825519 key (E/26-FAB0-9CC3-7F5E). Each line references a coordinator-signed ledger entry. Changing any

+4 hours forged signature breaks instantly



Impact & opportunity deserts

lowa · 99 counties · anonymized, k ≥ 10

<10

People supported (live)

<10

Recognized as exempt

<10

Covered, on a plan

4

Volunteer hours verified

Where 80 hours are out of reach without a car

Opportunity desert · 64

Tight · 18

Enough hours · 17

64

Deserts

64 of 99 counties

are opportunity deserts: car-free accessible volunteer hours cover less than half of what people subject to the rule need. Tap a county.

County data (CSV)

Who uses this map



Food banks & pantries

Where to open the next host site



Funders & companies

Measurable impact in meals and dollars



Officials & Iowa HHS

Counties under strain, failure reasons



Researchers & press

Aggregated, downloadable data

Model: SNAP participants = population × Iowa rate (247,907 recipients, Iowa HHS May 2026); subject = 14% of participants; 35% need ~20 volunteer h/month. Supply: ~1 volunteer-hosting organization per 1,200 residents × 150 h/month × car-free share (60% with fixed-route transit, 40% mid-size, 25% small rural counties), plus registered host sites. The pilot replaces the baseline with the live site registry and transit data.

06 · For Iowa

64

of 99 counties

are **opportunity deserts**: car-free volunteer hours cover less than half of what people under the rule need.

Neutral data — where to open the next host site, or where to ask for a waiver. Never a cell under 10 people.



Count Me In

Hack Away Hunger 2026 · dsmHack × Corteva

Food aid preserved per year

\$303,318

for 500 people supported · every number editable on the live dashboard

$500 \times 40\%$ would lose SNAP $\times 75\%$ kept = **150 people**

$150 \times \$168.51$ Iowa average benefit $\times 12$ months

89,211

meals / year

36,000

volunteer hours / year

~\$5k

running cost / year

How it's built

AI understands. Rules decide. A human settles doubts.

The person

SMS · voice · web

The coordinator

QR · one-tap validation

Public & officials

Dashboard · map

Multilingual AI agent

Claude / OpenAI · structured outputs

Letter reader

Vision → summary + deadline

Rules engine

One YAML per state, cited

Matching

Walk · DART · schedule

Signed hours ledger

Hash chain · Ed25519 · audit

Human relay

Navigator console

Verifiable attestation

PDF + QR

Iowa HHS

Only after a YES

Anonymized data

$k \geq 10$ by county



Count Me In

(515) 800-8080 · Kiswahili



Jambo

Habari! Mimi ni Count Me In. Ninakusaidia kubaki na msaada wako wa chakula (SNAP) chini ya sheria mpya ya saa 80 — bure, kwa ujumbe au kwa simu.

Faragha kwanza: ninahifadhi jina lako la kwanza, simu, lugha na saa tu. Hakuna kinachoenda kwa Serikali bila YES (NDIYO) yako. Jibu DELETE (FUTA) wakati wowote kufuta kila kitu, au HELP (MSAADA) kuongea na mtu.
Ninazungumza: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

Jina lako la kwanza ni nani?



Text message



Send the sample State letter



Dignity · access · privacy

Any phone. Any language. Always a human.



8 languages + voice for people who read little



Any phone keywords & 4-digit codes work without AI



HELP, always a human navigator one word away



DELETE by SMS minimal data, pseudonymous ledger



One option never a condition to receive food

What's next

Oct 2026

MVP

Full SMS journey, QR, attestation, map

Nov 26 – Mar 27

DMARC pilot

3–5 host sites · English + Spanish

2027

All of Iowa

55 counties · 8 languages · HHS link

2028

Other states

One rules file per state · open source

We don't create food. **We stop it** from disappearing.



count-me-in-iowa.netlify.app

>>

Judging runs on your own AI key (temporary). If we win, the prize funds the keys for everyone.

Inspiration

I'm sixteen. I have never received a letter from a government agency that could take away my food. Linda has.

Linda is 58, lives in Des Moines, cleans houses ten hours a week and has no car. In October an envelope arrives. Two pages of administrative English, and somewhere in the middle, the sentence that matters: from now on she must show **80 hours a month** of work, training or volunteering, or her SNAP food assistance stops. She has 21 days.

Linda is a composite. I built her out of numbers, and the numbers are real. The 2025 federal law (P.L. 119-21) extended SNAP's work rule from age 54 to 64, kept the parent exemption only for households with a child under 14, and removed the exemptions for veterans, people experiencing homelessness and young adults who aged out of foster care. In one year, Iowa's SNAP rolls fell **9%**, from 271,880 to 247,907 people. Meanwhile **1 in 8 Iowans** don't know where their next meal is coming from, and the pantry lines in Des Moines keep breaking records. When someone loses SNAP, they don't disappear. They join that line.

I started where I always start: I read everything, then I wrote it down. Before a single line of code existed, I had a 21-page brief: personas, objections, a demo script, the roadmap.

On page three of my own notes, I found the sentence that became the project. It wasn't in the news. It was in the regulation itself, **7 CFR 273.24**: volunteering with a public, faith-based or community organization **already counts toward the 80 hours, hour for hour**, as long as the hours are verified.

So the law already gives Linda a way to keep her food. What it doesn't give her is a way to prove she used it.

That changed the whole problem. People don't lose SNAP because they refuse to work. They lose it because they don't know whether they're exempt, they can't read the letter, they can't reach hours without a car, they can't prove what they did, or they miss one deadline. And two streets away, the same pantries are short of regular weekday volunteers and still track attendance on a paper sign-in sheet.

One well-recorded hour helps three people at once. **Linda keeps her food. The pantry gains a volunteer. The State gets clean proof.**

In maths class, when a proof lands, I shout **"C'est du génie !"** out loud. My teacher has given up trying to stop me. This idea felt like that kind of proof: invisible before you see it, obvious after.

Why a sixteen-year-old

I can't vote on this law. I can't sign a lease, and I don't run a food bank. What I can do is notice when the hard part of a problem isn't the hard part everyone is looking at.

I taught myself English with an app, one day at a time, until one morning a video made complete sense. A few years ago I stopped learning to code and studied something else: who a product is for, and how to explain it to them. I still don't know if that was smart or lazy. But it's why Count Me In starts with a woman holding a letter,

not with a database. It's also why every screen asks the same question: would Linda understand this, on a cheap phone, tired, in her second language?

■ What it does

Count Me In turns volunteering into verified hours.

For the person, on any phone, in 8 languages. No app, no account, no password. She texts a number, calls it, or uses the web.

- **The letter, decoded.** She photographs the letter, crooked and badly lit. She gets back three plain sentences: what it says, the deadline, and the one thing to do. A reminder is set automatically.
- **"Does this apply to me?"** Six questions, one at a time, answered in her own words. "My knees hurt a bit but I can still work." A deterministic rules engine, not the AI, decides between "probably exempt", "covered" and "let's check with a human". It cites the regulation every time.
- **A plan.** An 80-hour gauge, then **three missions at most**, ranked by real access: a 12-minute walk, a DART bus line, her cleaning schedule, her language, her knees.
- **Proof.** She checks in at the door by scanning a **QR code**, or by texting a **4-digit code** from any basic phone.
- **Control.** HELP reaches a human navigator. DELETE erases everything.

For host sites. Pantries, churches, gardens and corporate volunteer programs sign up in about ten minutes and print a QR poster. The coordinator sees a first name and a time slot, never a SNAP file. They confirm hours in one tap, and each confirmation is signed with the site's Ed25519 key. At the end of the month they get a report ready to attach to a grant application.

For the State. Every month, Linda receives an attestation: a PDF with a QR code. A caseworker scans it and knows in one second that it's authentic. **Change a single digit, and the signature breaks.** Nothing goes to Iowa HHS without her explicit YES, and that YES is logged.

For Iowa. A public, anonymized dashboard (no number ever describes fewer than 10 people), and a **map of opportunity deserts**: for each of the 99 counties, how many volunteer hours are reachable without a car compared with how many are needed. Under our model, 64 counties fall short, most of them rural. It's the kind of data that tells a food bank where to open its next site, and tells a policymaker where the rule can't physically be met.

Impact you can check in your head: 500 people supported × 40% who would lose SNAP × 75% who keep it thanks to the tool = 150 people. 150 × \$168.51 (Iowa's real average benefit, May 2026) × 12 months ≈ **\$303,000 of food assistance preserved per year**, plus about 36,000 volunteer hours for host sites. The running cost is a few thousand dollars a year.

■ How we built it

I built Count Me In with an AI coding agent. My job was the brief, the architecture, the decisions, and above all saying "no, that's not it" until it was.

- **React + TypeScript + Vite**, mobile-first and accessible: large touch targets, high contrast, right-to-left Arabic, a Burmese font, and a voice mode (Web Speech API) for people who read little.
- **One agent for every channel.** The same code answers the web phone and the **Twilio** SMS and voice webhooks. Keywords and site codes are handled first, without any AI, so a basic phone or an outage never breaks the flow.
- **The AI layer:** Claude (Anthropic SDK) or OpenAI, always with **structured JSON outputs**. It does three things only: understand free text, read letter photos (vision + OCR), and translate.
- **The rules engine:** `rules/iowa.yaml`, one readable file per state, with its sources. When the law changes, you edit a file a paralegal can read, not the code.
- **The signed ledger:** append-only and hash-chained, with Ed25519 signatures (TweetNaCl). It stores a salted pseudonym, never a phone number.
- **Attestations:** signed on the server, rendered with **pdf-lib**, with a QR code and a verification page that also works by camera (**jsQR**).
- **The map:** my own SVG renderer of Iowa's county boundaries. No third-party map tiles.
- **Quality: Vitest** (20 tests, including forgery and tampering cases) and a full **Playwright** end-to-end run. Hosted on **Netlify**.

About the API keys (temporary). For judging, the app asks you for your own Claude or OpenAI key. It stays in your browser and is never in the code. **If we win, the prize will fund the AI, SMS and voice keys so the service runs for everyone.** People like Linda will never see a key. They'll just text a number. Because I shipped no keys, the demo video uses pre-written AI responses; the live app runs on the real APIs.

■ Challenges we ran into

- **The law didn't agree with itself.** Early drafts of the bill set the parent threshold at a child under 7; the final law says under 14. I decided to encode only what I could cite, and to put every source in the rules file.
- **Keeping the AI out of decisions without making it cold.** The AI is only allowed to turn words into facts; the rules engine decides. Making that still sound like a kind person took many rounds.
- **Translation that doesn't break SMS.** "Reply 1, 2 or 3", a 4-digit code and the word HELP all have to survive Swahili and Arabic untouched.
- **Proof that is actually hard to fake.** I wrote tests that try to forge an attestation, turning 4 hours into 8, and made sure every one of them fails.
- **I hated my first design.** The first version worked and looked generic. I threw away the entire interface and rebuilt every screen, the slides and the video around one design system. The product works the same. It just finally looks like something Linda would trust.
- **Staying honest.** I used Iowa's real average benefit, \$168.51, instead of a rounder number, even though it made my headline smaller.

■ Accomplishments that we're proud of

A complete journey you can test in two minutes, from "Hi" to a signed attestation a caseworker can trust in one second. A rules file a paralegal can read. A product that works on any phone, in eight languages, with a human always one word away.

And the thing I'm proudest of is a sentence that isn't in the code: **the problem is proof, not willingness.** Once I understood that, every design decision followed from it.

■ What we learned

The bottleneck is verification, not willingness. Most of this problem is a paperwork problem, and paperwork is something software is good at.

"AI-first" doesn't have to mean "AI decides". Putting a cited rules engine between the model and the outcome made the tool easier to trust, and easier to explain to a pastor or a caseworker.

Dignity has to be designed in. We say "missions" and "hours that count", never "control" or "sanctions". A Count Me In volunteer looks exactly like every other volunteer, and nothing leaves without a yes.

Being sixteen is not a reason to build something small. It's a reason to read the regulation more carefully than the people who assume they already know it.

■ What's next for Count Me In

1. **A pilot in the DMARC pantry network** (Nov 2026 – Mar 2027): 3 to 5 host sites, English and Spanish first.
2. **Validating the attestation format with Iowa HHS.**
3. **Scaling to Food Bank of Iowa's 55 counties**, then to other states, one rules file each, because the law is federal and every state has the same problem.

Along the way: move the ledger to a server-side database, load DART's full transit feed, and fund the keys. The code is open source (MIT), so a nonprofit can host it and dsmHack volunteers can maintain it. I'd love Corteva to be the first corporate host site.

I can't change the law. I can make sure nobody loses their food because they couldn't prove an hour they actually gave.

We don't create food. We stop it from disappearing.

Demo host sites, people and the sample letter are illustrative; no partnership is implied.